STEPHEN KURL G. PINACATE



Computer Engineering Student | Software Engineering & Full-stack Development

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PROFESSIONAL OBJECTIVE

Adaptable and results-driven Computer Engineering student and DOST scholar with hands-on experience in full-stack software development, mobile applications, and embedded systems. Experienced in designing, developing, and deploying end-to-end solutions, including systems actively used in business operations. Proficient in C#, ASP.NET Core, SQL Server, and modern development technologies, with strong analytical and problem-solving skills. Committed to delivering reliable, maintainable solutions, adapting quickly to new technologies, and contributing effectively to collaborative engineering teams.

EDUCATION

Angeles University Foundation — Bachelor of Science in Computer Engineering 2023 – Present

- University Scholar (2024–Present); reported general weighted average: 95.
- DOST Scholar (2023–Present); College Scholar (2023–2024), reported general weighted average: 90.

SKILLS AND COMPETENCIES

Programming languages: C#, C, C++, Python, JavaScript/TypeScript, SQL

Web and backend development: ASP.NET Core, REST APIs, JWT authentication, PHP, React.js, HTML/CSS

Desktop and mobile: WinUI 3, .NET, Kotlin, Android Jetpack Compose, React Native

Databases and persistence: SQL Server, PostgreSQL, MySQL, MongoDB, Supabase, Entity Framework Core, SQLite

Engineering workflow: Git/GitHub, Agile collaboration, role-based access, offline-first synchronization concepts

HARDWARE AND EMBEDDED SYSTEMS

Relevant foundation: C/C++ programming, circuit construction, breadboarding and electronics troubleshooting through Computer Engineering coursework and regional competition participation.

Technical participation: 9th Regional CpE Challenge (2026) — Breadboarding Competition participant and 2nd Place in the Networking Competition; Cisco APJC NetAcad Riders Round 1 (2026), focused on networking and technical troubleshooting.

FREELANCE / WORK EXPERIENCE

INNOVET — Multi-Branch Veterinary Clinic Management Platform Solo | Full-stack Developer

Engineered a scalable veterinary operations platform designed to unify appointment scheduling, patient histories, client records, and branch administration across a planned network of 10 clinics and one veterinary hospital.

- Structured client, branch-admin, and super-admin workflows with controlled access and database-backed appointment and record handling; integrated across the clinic management platform.

Technologies: ASP.NET Core, C#, Entity Framework Core, SQL Server.

VITALLY — Clinic Operations and Transaction Platform Solo | Full-stack Developer

Independently delivered a production-used clinic management platform serving 2 clinics, consolidating financial transactions, inventory control, expense tracking, reporting, and audit records into a single mobile-friendly system.

- Significantly improved day-to-day clinic workflows by centralizing transactions, inventory management, and operational reporting, reducing reliance on fragmented manual processes.

Technologies: ASP.NET Core, SQL Server, Responsive Web/PWA.

SELECTED SOFTWARE ENGINEERING PROJECTS

TUKI — AI-Assisted Public Transport Navigation Team | Full-stack / Mobile Developer | GitHub

Helped develop an integrated public transportation navigation platform that guides commuters through realistic, multi-leg journeys involving walking, tricycles, and jeepneys across Pampanga.

- Worked on routing and map behavior, backend integration, live trip state, fares, transfers, navigation guidance, and rerouting.

Technologies: Kotlin, Jetpack Compose, ASP.NET Core, SQL Server.

AUFTHORIZE — Digital Request and Approval System Two-developer team | Full-stack Developer | GitHub

Co-developed a digital approval platform that streamlines institutional requests—from submission to multi-level review—for venues, budgets, liquidation, reimbursements, and materials.

- Designed multi-stage approval behavior, role-based review, e-signature processes, document generation, attachments, and email notifications.

Technologies: PHP, MySQLi, PDF Generation, PHPMailer.

ENGRLINK — Academic Portal Prototype Three-developer team | Desktop / Full-stack Developer | GitHub

Co-developed a role-based academic management prototype that brings together student records, grading, scheduling, fee monitoring, and administrative services for the AUF College of Engineering and Architecture.

- Supported enrollment, scheduling, grades, balances, subject offerings, role-based interfaces, and academic-data workflows.

Technologies: C#, WinUI 3, .NET, Supabase.

GAUSS BUSTERS — Numerical Methods Desktop Solver Solo | Desktop Developer | GitHub

Independently built an interactive numerical computing application that transforms complex mathematical problems into traceable, step-by-step solutions with visual representations of results.

- Implemented Bisection, Newton–Raphson, Secant, Gaussian Elimination, regression, step-by-step iteration results, and numerical plotting.

Technologies: C#, .NET 8, WinUI 3, ScottPlot.

PORTFOLIO SITE — Personal Developer Portfolio | 2026 | Portfolio

Designed and deployed a responsive developer portfolio that presents practical software and embedded engineering work, technical competencies, and achievements through an accessible professional showcase.

Technologies: Next.js, React, TypeScript, Tailwind CSS.

CURRENT PROJECTS AND DEVELOPMENT

AUF NAVI VISION — Context-Aware Campus Information & Navigation Individual Thesis | In Development

Developing a context-aware campus platform that connects verified institutional information with location-aware guidance to university offices and facilities.

- Designing shared campus-entity workflows linking cited, authority- and time-aware RAG with GPS outdoor navigation and vision/OCR-assisted checkpoint-based indoor routing using A*.

Technologies: ASP.NET Core, .NET 10, SQL Server; planned Kotlin/Jetpack Compose, OpenAI API/RAG, CameraX, ML Kit OCR, MapLibre.

ICpEP REGION 3 HUB — Regional Chapter Management Website Solo Full-stack Developer | In Progress | GitHub

Independently developing a regional coordination platform designed to connect ICpEP.se chapters through centralized announcements, event management, chapter directories, assistance requests, and collaboration tools.

- Implementing role-based chapter and regional workflows, including account management and an administrative portal; development is ongoing.

Technologies: Next.js, React, NestJS, Prisma, PostgreSQL.

SMARTFARM — Smart Farm & Crop Monitoring Platform Collaborative Project | Web + Android | GitHub

Co-developing a web-and-Android farm-management platform designed to help farm owners and field teams manage multiple farms, plots, crop cycles, daily field activities, supply inventory, harvest records, and weather-related reminders.

- Designed around role-based farm access, crop and activity monitoring, inventory and partial harvest recording, with selected Android workflows intended for offline use and synchronization when connectivity returns.

Technologies: .NET 10 / ASP.NET Core, EF Core 10, SQL Server 2022, Hangfire, Kotlin / Jetpack Compose, Room, Retrofit / OkHttp, WorkManager.

BE MY PAROL — Interactive Digital Christmas Lantern Platform Three-developer team | Full-stack Developer

Co-developing a collaborative holiday experience where users create personalized digital lanterns, invite friends to contribute without registering, and reveal collected messages during the Christmas celebration.

- Building account authentication, guest contribution workflows, database-backed lantern designs, and responsive SVG-based interfaces.

Technologies: React, TypeScript, Vite, ASP.NET Core (.NET 10), C#, SQL Server, Entity Framework Core, Tailwind CSS.

APPLIED HARDWARE AND TECHNICAL PROBLEM-SOLVING

ESP32 Sumobot Controller — Academic Prototype Embedded / Mobile Integration

Developed and refined a mobile sumobot prototype that combines remote control with sensor-assisted movement, opponent detection, and boundary awareness for competitive robotic behavior.

- Investigated motor direction and power behavior, sensor readings, intermittent sensor failures, and manual versus autonomous navigation logic. Iterated on robot control logic, sensor integration, and testing.

Technologies: ESP32, Flutter, Motor Driver, Distance and Edge Sensors.

Spider Robot – Multi-Legged Embedded Robotic System | In Progress, 2026

Developing a multi-legged robotic prototype focused on coordinated servo-driven movement, stable directional control, and sensor-responsive behavior as a foundation for autonomous mobility.

- Designed to support directional movement, obstacle detection, and sensor-based responses, with potential manual or autonomous control.

Technologies: ESP32 (proposed), C/C++, PWM, GPIO, Servo Motors, Sensors.

Smart Environmental Monitoring System with Flutter Dashboard | 2026

Built an end-to-end environmental monitoring prototype that makes live temperature and humidity measurements accessible remotely through a mobile dashboard.

- Implemented periodic data updates and validated sensor readings against the dashboard, with a temperature-triggered buzzer alert specified as a future extension.

Technologies: ESP32, C/C++, Flutter, Dart, DHT22, HTTP, Wi-Fi, REST-style API.

Embedded Motor and Actuator Control System with Flutter Interface | 2026

Built a wireless actuator-control prototype that brings precise servo positioning, bidirectional stepper movement, and adjustable output control into one mobile-operated interface.

- Implemented wireless actuator commands through a mobile interface, supporting servo positioning, bidirectional stepper movement, output adjustment, and system reset.
- Integrated analog input monitoring and real-time actuator status reporting, validating hardware responses against mobile application commands.

Technologies: ESP32, C/C++, Flutter, Dart, PWM, GPIO, ADC, HTTP, Wi-Fi, Servo Motors, Stepper Motors.

Smart Sensor–Actuator Automation System with Flutter Control | 2026

Designed an automated sensing-and-response prototype that combines environmental monitoring and motion detection to trigger appropriate mechanical and audible actions, with manual override through a mobile dashboard.

- Implemented sensor-based decision logic for automatic actuator responses, including motion-triggered servo operation and temperature-based alerts.
- Built a mobile dashboard for real-time sensor monitoring and switching between manual and automatic control modes.

Technologies: ESP32, C/C++, Flutter, Dart, DHT11, PIR Sensor, Servo Motors, GPIO, PWM, Wi-Fi, HTTP.

Digital Door Lock System Using Logic Circuits | 2025

Designed and constructed a configurable hardware-only digital security prototype capable of setting, changing, and validating access passwords entirely through interconnected logic circuitry.

- Implemented hardware-based password setting, modification, and verification through combinational and sequential logic circuits.

Technologies: Digital Logic ICs, Comparators, Flip-Flops, DIP Switches, Breadboard Circuit Design, Combinational and Sequential Logic.

Spinergy – Rotary-Controlled Power Supply | 2025

Designed, assembled, and tested a practical multi-output DC power supply delivering selectable voltage levels of approximately 1.25V, 3V, 6V, 9V, and 12V for electronics experimentation.

- Implemented resistor-based voltage regulation, capacitor filtering, LED voltage indicators, and manual power switching using a soldered PCB assembly.
- Performed circuit testing and voltage measurements using a multimeter to verify output levels and functionality.

Technologies: LM317T, Analog Circuit Design, Voltage Regulation, PCB Assembly, Soldering, Multimeter Testing.

ACHIEVEMENTS AND DISTINCTIONS

- 2026 — Overall Champion, 9th Regional CpE Challenge (member of the recognized delegation).
- 2026 — 2nd Place, Networking Competition, 9th Regional CpE Challenge.
- 2024–Present — University Scholar, Angeles University Foundation (reported 95 GWA).
- 2023–Present — Department of Science and Technology (DOST) Scholar.
- 2023–2024 — College Scholar, Angeles University Foundation (reported 90 GWA).
- Most Outstanding Student Council — First Place (team recognition).
- Eureka Award — Member of an organization recognized for academic excellence.
- 2019–2020 — Merit Award, Philippine International Mathematical Olympiad (PhiMO).
- 2018–2019 — Bronze Award, Thailand International Mathematics Olympiad (TIMO).

LEADERSHIP AND ORGANIZATION EXPERIENCE

Vice President – Internal — ICpEP.se Region 3 | 2026 – Present

President — ICpEP.se — AUF Chapter | 2026 – Present

Vice President — AUF College of Engineering and Architecture Student Council | 2025 – 2026

Treasurer — ICpEP.se — AUF Chapter | 2025 – 2026

Junior Officer — AUF College of Engineering and Architecture Student Council | 2024 – 2025

Technical Coordinator — CEA Athletics and Talents Support Squad | 2024 – 2025

Second Year Representative — ICpEP.se — AUF Chapter | 2024 – 2025

Class Vice President — Class Organization | 2023 – 2024

COMPETITIONS AND TECHNICAL PARTICIPATION

- 2026 — Can You HackIT? The IBPAP Challenge (Clark): Hackathon Finalist (team); Best School Award (school recognition).
- 2026 — Cisco APJC NetAcad Riders, Round 1 participant; networking concepts, troubleshooting and technical problem-solving.
- 2026 — 9th Regional CpE Challenge: 2nd Place in the Networking Competition and participant in the Breadboarding Competition.

PROFESSIONAL REFERENCES

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